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Glucose in Urine Test

What is a glucose in urine test?

A glucose [<https://medlineplus.gov/bloodglucose.html>] in urine test measures the amount of glucose in your urine (pee). Glucose is a type of sugar. It is your body's main source of energy. Glucose comes from the food you eat. Your body breaks down most of that food into glucose and releases it into your bloodstream. Your blood carries glucose to your cells for energy.

Normally, urine contains very little or no glucose. That's because your kidneys hold on to glucose and send it back to your bloodstream instead of flushing it out of your body in urine. If you have more glucose than normal in your urine, it could mean that you have:

- Too much glucose in your blood, which could be a sign of **diabetes** [<https://medlineplus.gov/diabetes.html>] . Diabetes is the most common cause of high urine glucose levels.
- A rare kidney condition that causes your kidneys to remove glucose from your blood even when your blood glucose levels are normal. These conditions include:
 - Renal glycosuria, a genetic condition that is inherited (passed down through families)
 - Fanconi syndrome, a genetic condition you can inherit or develop later in life (Fanconi syndrome is not related to Fanconi anemia [<https://medlineplus.gov/genetics/condition/fanconi-anemia/>])

If your urine glucose levels are high, your health care provider will usually order a **blood glucose test** [<https://medlineplus.gov/lab-tests/blood-glucose-test/>] to help make a diagnosis.

Other names: urine sugar test; urine glucose test; glucosuria test; glycosuria test

What is it used for?

A glucose in urine test is usually part of a **urinalysis** [<https://medlineplus.gov/urinalysis.html>] , a test that measures different cells, chemicals, and other substances in your urine. A urinalysis is often used to check your general health.

In the past, urine glucose testing was commonly used to screen for and monitor diabetes. But urine glucose testing is not as accurate as blood glucose testing, so blood tests are now more commonly used to diagnose and monitor diabetes.

A urine glucose test may still be used to help diagnose and monitor diabetes if you have a condition that makes blood testing difficult. For example, you may not be able to have a blood test if your veins are scarred from past blood tests or if you have a type of **bacterial infection** [<https://medlineplus.gov/bacterialinfections.html>] that could spread to your blood if a needle is inserted into a vein.

A glucose in urine test may also be done if your provider thinks you could have an uncommon type of kidney disorder.

Why do I need a glucose in urine test?

You may have a glucose in urine test as part of your **routine checkup** [<https://medlineplus.gov/healthcheckup.html>] . You may also need this test if you have symptoms of diabetes or need to monitor your diabetes, but you are unable to have a blood glucose test.

Symptoms of diabetes include:

- Increased thirst and urination
- Increased hunger
- **Fatigue** [<https://medlineplus.gov/fatigue.html>]
- **Blurred vision** [<https://medlineplus.gov/visionimpairmentandblindness.html>]
- Numbness or tingling in the hands or feet
- Sores that don't heal
- Losing weight without trying

If you are pregnant, you will likely have a glucose in urine test as part of a urinalysis to check your health. Some glucose in urine is common during pregnancy. But high levels over time could be a sign of **gestational diabetes** [<https://medlineplus.gov/diabetesandpregnancy.html>] , which is a type of diabetes that happens only during pregnancy [<https://medlineplus.gov/pregnancy.html>] .

You may need a glucose in urine test if your provider thinks you could have an uncommon kidney problem that affects how your kidneys filter glucose in blood. These conditions don't always cause symptoms. Your provider may want to test you because you have a family history of one of these conditions.

What happens during a glucose in urine test?

You will need to give a urine sample for the test. A health care professional may give you a cleansing wipe, a small container, and instructions for how to use the "clean catch" method to collect your urine sample. It's important to follow these instructions so that germs from your skin don't get into the sample:

1. Wash your hands with soap and water and dry them.
2. Open the container without touching the inside.
3. Clean your genital area with the cleansing wipe:
 - For a penis, wipe the entire head (end) of the penis. If you have a foreskin, pull it back first.
 - For a vagina, separate the labia (the folds of skin around the vagina) and wipe the inner sides from front to back.
4. Urinate into the toilet for a few seconds and then stop the flow. Start urinating again, this time into the container. Don't let the container touch your body.
5. Collect at least an ounce or two of urine into the container. The container should have markings to show how much urine is needed.
6. Finish urinating into the toilet.
7. Put the cap on the container and return it as instructed.

If you have hemorrhoids [<https://medlineplus.gov/hemorrhoids.html>] that bleed or are having your menstrual period [<https://medlineplus.gov/menstruation.html>] , tell your provider before your test.

Your provider may ask you to monitor your urine glucose at home with a test kit. Your **home test kit** [<https://medlineplus.gov/lab-tests/at-home-medical-tests/>] will come with a package of strips for testing a sample of your urine and instructions on how to do the test. Be sure to follow the instructions carefully. Talk with your provider about what time of day you should test your urine to get the most accurate results.

Will I need to do anything to prepare for the test?

You usually don't need any special preparations for this test. Some medicines and supplements may affect your test results, so be sure to tell your provider about everything you're taking. But don't stop taking any medicine without talking with your provider first.

Are there any risks to the test?

There is no known risk to having a glucose in urine test.

What do the results mean?

If your test result is normal, it means that you had little to no glucose in your urine. But this doesn't mean that you don't have diabetes. You can have a normal urine glucose level when your blood glucose level is high.

If your test result shows you have too much glucose in your urine, it may be a sign of:

- Diabetes (If you have already been diagnosed with diabetes, high urine glucose may mean that your diabetes is not well controlled)
- Pregnancy
- Gestational diabetes
- An uncommon type of kidney disorder

A urine glucose test alone cannot diagnose any conditions. So, if you have an abnormal test result, your provider may order other tests to help make a diagnosis, including **blood tests** to diagnose or rule out diabetes [<https://medlineplus.gov/lab-tests/diabetes-tests/>] .

Too much urine glucose doesn't always mean you have a medical condition that needs treatment. Certain medicines, vitamin C, and eating a large meal may affect your test results. Ask your provider what your test results mean and whether you need a blood glucose test for more accurate results.

Learn more about **laboratory tests**, **reference ranges**, and **understanding results** [<https://medlineplus.gov/lab-tests/how-to-understand-your-lab-results/>] .

References

1. American Diabetes Association [Internet]. Arlington (VA): American Diabetes Association; c1995-2024. Check Your Blood Glucose: Diabetes Testing & Monitoring; [cited 2024 Nov 20]; [about 6 screens]. Available from: <https://www.diabetes.org/healthy-living/medication-treatments/blood-glucose-testing-and-control/checking-your-blood-sugar> [<https://www.diabetes.org/healthy-living/medication-treatments/blood-glucose-testing-and-control/checking-your-blood-sugar>]
2. American Pregnancy Association [Internet]. Irving (TX): American Pregnancy Association; c2024. Getting A Urinalysis: About Urine Tests; [cited 2024 Nov 20]; [about 3 screens]. Available from: <https://americanpregnancy.org/prenatal-testing/urine-test-urinalysis/> [<https://americanpregnancy.org/prenatal-testing/urine-test-urinalysis/>]
3. Cleveland Clinic: Health Library: Diseases and Conditions [Internet]. Cleveland (OH): Cleveland Clinic; c2024. Glycosuria; [reviewed 2024 May 17; cited 2024 Nov 20]; [about 12 screens]. Available from: <https://my.clevelandclinic.org/health/diseases/glycosuria> [<https://my.clevelandclinic.org/health/diseases/glycosuria>]
4. Cleveland Clinic: Health Library: Diseases and Conditions [Internet]. Cleveland (OH): Cleveland Clinic; c2024. Fanconi Syndrome [reviewed 2022 Sep 28; cited 2024 Nov 20]; [about 18 screens]. Available from: <https://my.clevelandclinic.org/health/diseases/24233-fanconi-syndrome> [<https://my.clevelandclinic.org/health/diseases/24233-fanconi-syndrome>]
5. Keefe P, Bokhari SRA. Fanconi Syndrome. [Updated 2023 Jul 4; cited 2024 Nov 20]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK534872/> [<https://www.ncbi.nlm.nih.gov/books/NBK534872/>]
6. Liman MNP, Jialal I. Physiology, Glycosuria. [Updated 2023 Mar 13; cited 2024 Nov 20]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK557441/> [<https://www.ncbi.nlm.nih.gov/books/NBK557441/>]

7. Merck Manual Consumer Version [Internet]. Kenilworth (NJ): Merck & Co., Inc.; c2026. Renal Glucosuria [2026 Mar; cited 2026 Jul 06; [about 2 screens]. Available from: <https://www.merckmanuals.com/home/kidney-disorders/disorders-of-kidney-tubules/renal-glucosuria>
8. Merck Manual Consumer Version [Internet]. Kenilworth (NJ): Merck & Co., Inc.; c2024. Urinalysis and Urine Culture [2024 Jan; cited 2024 Nov 20]; [about 4 screens]. Available from: <http://www.merckmanuals.com/home/kidney-and-urinary-tract-disorders/diagnosis-of-kidney-and-urinary-tract-disorders/urinalysis> [http://www.merckmanuals.com/home/kidney-and-urinary-tract-disorders/diagnosis-of-kidney-and-urinary-tract-disorders/urinalysis]
9. National Cancer Institute [Internet]. Bethesda (MD): U.S. Department of Health and Human Services; NCI Dictionary of Cancer Terms: glucose [cited 2024 Nov 20]; [about 1 screen]. Available from: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/glucose> [https://www.cancer.gov/publications/dictionaries/cancer-terms/def/glucose]
10. National Institute of Diabetes and Digestive and Kidney Diseases [Internet]. Bethesda (MD): U.S. Department of Health and Human Services; Gestational Diabetes; [reviewed 2017 May; cited 2024 Nov 20]; [about 2 screens]. Available from: <https://www.niddk.nih.gov/health-information/diabetes/overview/what-is-diabetes/gestational> [https://www.niddk.nih.gov/health-information/diabetes/overview/what-is-diabetes/gestational]
11. Srikanth KK, Lotfollahzadeh S. Phlebotomy. [Updated 2023 Aug 28; cited 2024 Nov 20]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK574569/#> [https://www.ncbi.nlm.nih.gov/books/NBK574569/#]
12. Testing.com [Internet]. Seattle (WA): OneCare Media; c2024. Glucose Testing; [modified 2021 Nov 9; cited 2024 Nov 20]; [about 12 screens]. Available from: <https://www.testing.com/glucose-testing/> [https://www.testing.com/glucose-testing/]
13. Testing.com [Internet]. Seattle (WA): OneCare Media; c2024. Tips on Blood Testing; [modified 2021 Jan 27; cited 2024 Nov 20]; [about 5 screens]. Available from: <https://www.testing.com/articles/laboratory-testing-tips-blood-sample/> [https://www.testing.com/articles/laboratory-testing-tips-blood-sample/]
14. Testing.com [Internet]. Seattle (WA): OneCare Media; c2024. Urinalysis; [modified 2022 Sep 28; cited 2024 Nov 20]; [about 11 screens]. Available from: <https://www.testing.com/tests/urinalysis/#glucose> [https://www.testing.com/tests/urinalysis/#glucose]

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